

A major limitation of performing statistical analysis using GradCafe is that the dataset is produced through anonymous, voluntary, self-reporting rather than through a controlled sampling process. This means that both the decision to appear in the dataset and the information contained in each record are subject to selection and self-reporting biases. An applicant must know about GradCafe, visit the website, choose to submit a result, and then decide which pieces of information (such as GPA, GRE scores, degree type, citizenship status, and comments) to disclose. Applicants who complete this process are not necessarily representative of the broader population of graduate applicants. For example, applicants with particularly positive outcomes may be excited to report an acceptance, while applicants with particularly negative or surprising outcomes may post because they want to share an unexpected rejection or compare their experience with other applicants. Applicants with ordinary outcomes may have less motivation to post at all. The same self-selection can occur within individual fields: someone with a very high GRE score or GPA may be more willing to include it than someone with a lower score. As a result, removing NULL values before calculating an average solves the computational problem of missing data, but it does not solve the statistical problem created by the missingness. If applicants who omit their GPA or GRE scores systematically differ from those who report them, the average among the reported values can be biased even when the SQL query calculating that average is completely correct.

The reliability of the reported values themselves presents an additional and observable problem in this dataset. During my analysis of GRE scores, I initially obtained averages that were impossible under the actual GRE scoring scales. Inspection of the underlying records revealed reported GRE Quantitative or Verbal values such as **999, 990, 980, and 800**, despite the relevant GRE section scores being expected to fall within the **130–170** range. The data also contained Analytical Writing values greater than the valid **0–6** range. These were not merely small measurement errors that would have an insignificant effect on an average; values several times larger than the legitimate maximum can substantially distort a mean, particularly if they are not detected before aggregation. For this reason, my GRE analysis ultimately restricted Quantitative and Verbal scores to values between **130 and 170** and Analytical Writing scores to values between **0 and 6**. After applying these validity restrictions, my database produced averages of **165.74 for GRE Quantitative, 160.37 for GRE Verbal, and 4.33 for GRE Analytical Writing**. However, even these cleaned averages must be interpreted cautiously. Filtering removes values that are obviously impossible, such as 999, but it cannot identify a fabricated or incorrectly entered value that happens to fall inside the legitimate range. For example, an anonymously submitted score of 168 is numerically plausible, so there is no way to determine from the database alone whether the applicant actually earned a 168. This illustrates an important distinction between data validation and data verification: I can identify some values that are impossible based on the allowable range, but I cannot independently verify that every plausible value is truthful or accurate. Therefore, cleaning improves the mathematical quality of the analysis without eliminating the fundamental reliability problem associated with anonymous self-reporting.

The same concerns affect statistics that appear completely reasonable, including my analysis of GPA among accepted Princeton University applicants for Fall 2026. My SQL and SQLAlchemy analyses found an average GPA of **3.87** among the qualifying accepted Princeton records for which GPA was reported. Unlike the obviously invalid GRE values, there is nothing inherently implausible about a 3.87 average for this subset of records. Nevertheless, the fact that a statistic is plausible does not make the underlying sample representative. The 3.87 value answers a narrow question: **what is the arithmetic mean of the**

non-missing GPA values among records in my GradCafe database that were identified as accepted Princeton Fall 2026 applicants? It does not answer the much broader question of what the average GPA was for every graduate applicant admitted to Princeton for Fall 2026. An admitted Princeton applicant with a 3.95 GPA may be particularly comfortable publishing that information, while another admitted applicant with a substantially lower GPA may report the acceptance but leave the GPA field blank. A third applicant may never use GradCafe, meaning that neither the admission outcome nor the GPA would appear anywhere in my database. The first applicant contributes to the calculated mean, the second may contribute to other analyses but not the GPA calculation, and the third is completely invisible. Consequently, the database provides no basis for assuming that the observed GPA values are a random sample of all accepted Princeton applicants. The missing values are especially important because calculating **AVG(gpa)** while excluding NULL values implicitly bases the result only on people who chose to provide a GPA; it cannot compensate for the unknown GPAs of people who did not report one.

Representativeness is also limited by differences in who uses GradCafe and which universities and programs generate the most submissions. The dataset does not contain a predetermined number of observations sampled proportionally from each university, degree type, academic discipline, domestic/international group, or admissions outcome. Instead, its composition is determined by user participation. Some academic communities may use GradCafe extensively, while applicants in other disciplines may rarely use it. Even within Princeton, the records included in my 3.87 calculation may represent an uneven mixture of programs with substantially different admissions practices and applicant populations. If applicants from one Princeton department are particularly active on GradCafe, their GPAs could have a disproportionate influence on an institution-level average. Similarly, applicants who are unfamiliar with GradCafe, do not participate in online admissions communities, do not wish to disclose academic information publicly, or simply have no interest in reporting their results are systematically absent from the data collection process. These absent applicants cannot be treated as if they had an equal chance of being sampled because GradCafe does not use random sampling in the first place. The dataset can therefore support descriptive statements about the records that were actually submitted (for example, that the qualifying accepted Princeton Fall 2026 records with reported GPAs in my database had an average GPA of 3.87) but it cannot establish that **3.87 is the true average GPA of Princeton's complete Fall 2026 admitted graduate population**. More generally, my analyses describe patterns within a large collection of voluntarily submitted GradCafe records. The size of that collection does not eliminate selection bias, self-reporting bias, missingness, measurement error, or unequal representation. A dataset can contain tens of thousands of observations and still provide a biased picture of the population if the mechanism determining who enters the dataset is non-random. The distinction between **what the database mathematically contains** and **what the database allows me to conclude about the real-world applicant population** is therefore essential when interpreting every result produced by this analysis.